# Exercise is Medicine for Concussion

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**Abstract**

Sport-related concussion (SRC) is a physiological brain injury that produces cerebral and systemic effects, including exercise intolerance. Exercise intolerance after concussion is believed to be the result of autonomic nervous system (ANS) dysfunction. Ventilation is inappropriately low for the level of exercise intensity, raising arterial carbon dioxide ($PaCO_2$) levels. Elevated $PaCO_2$ increases cerebral blood flow (CBF) out of proportion to exercise intensity, which is associated with symptoms that limit exercise performance. Thus, elevated exercise $PaCO_2$ may signal incomplete recovery from SRC. This article reviews recent observational and experimental data and presents the evidence that subthreshold aerobic exercise normalizes the cerebrovascular physiological dysfunction and is ''medicine'' for patients with concussion and persistent postconcussive symptoms (PPCS). It discusses the systematic evaluation of exercise tolerance after concussion using the Buffalo Concussion Treadmill Test (BCTT) and reviews the utility of the Buffalo Concussion Bike Test (BCBT), the data from which are used to establish an individualized heart rate ''dose'' of subthreshold exercise to safely speed recovery, which also may work in the acute recovery phase after SRC with the potential to reduce the incidence of PPCS. Evaluation and treatment approaches based on the physiology of concussion suggest that exercise is medicine for concussion, potentially adding a new dimension to concussion care to help safely speed recovery and prevent PPCS in some patients.

## Introduction

Sport-related concussion (SRC) is a traumatic physiological brain injury having widespread cognitive, emotional, and physical effects (1). One effect of concussion is exercise intolerance, that is, the inability to exercise to the level predicted for one's age and fitness (2). Concussed patients stop exercise because of symptom exacerbation, which appears to be an objective indicator or biomarker of ongoing physiological dysfunction after concussion (2–4). Most children and adolescents recover spontaneously within 2 to 4 wk after SRC (5,6) but approximately 30% experience persistent postconcussive symptoms (PPCS), defined as symptoms beyond 1 month after injury (7–9). Persistent postconcussive symptom negatively impacts quality of life in children and adolescents, including school and athletic performance (10,11).

There are no evidence-based therapies for SRC. Past consensus-based recommendations emphasized physical and cognitive rest until complete symptom resolution before return to training or sport (12). The concept of "rest is best" was supported by animal and human evidence of a vulnerable period early after concussion during which the brain is susceptible to repeat injury and/or worsening symptoms with cognitive or physical stress (13) plus evidence that *excessive activity* soon after concussion prolonged recovery (14,15). The concept of using rest until symptom resolution also was applied to those with PPCS. Recent evidence from a randomized controlled trial (RCT) showed, however, that strict rest beyond 2 d prolonged symptomatic recovery from concussion (16), consistent with the observation that removing athletes from regular physical activity is detrimental to their mental health (17). Recent observational data suggest that *moderate levels of physical activity* within the first week after injury reduces PPCS incidence in children and adolescents when compared with subjects who reported doing no structured physical activity (18). The most recent Concussion in Sport Group (CISG) international consensus statement, published in 2017, says that there is insufficient evidence that prescribing complete rest promotes recovery from concussion (1).

There are extensive data to suggest that exercise, specifically aerobic exercise, attenuates cognitive impairment and reduces dementia risk in humans (19). Proposed mechanisms

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include induction of factors that promote neuron growth and repair (such as brain-derived neurotrophic factor [BDNF], which increases hippocampal volume and improves spatial memory (20)) as well as mitigation of vascular disease risk. Animal studies confirm that exercise increases BDNF levels (19). Emerging human data suggest that exercise favorably affects brain neuroplasticity by increasing BDNF, which has been reported as early as 5 to 6 wk after initiation of aerobic training (21,22). The rapidity of the beneficial effect of exercise on neuroplasticity suggests that improved neuronal function rather than reduced vascular disease risk is most germane to the effect of exercise in concussion. The principle of "exercise is medicine" is that health care systems might begin to think of exercise as a medication that should be prescribed to patients (23). As with medication, however, it is essential that exercise be prescribed based on a "dosage" that suits the characteristics of the individual. The purpose of this article is to review the evidence for the use of controlled aerobic exercise as medicine for the treatment of concussion and PPCS.

### Aerobic Exercise and Concussion

Early studies showed that aerobic exercise imposed upon rodents within 2 wk of fluid percussion-simulated concussion was detrimental to neurotrophic factor expression and to recovery of cognitive function (14,24). Exercise administered, however, three or more weeks after injury was beneficial to both. Other animal studies showed that the motivation for exercise after brain injury is important. Rats *forced* to exercise after simulated concussion markedly stimulated the corticotrophic axis, increased stress hormone levels, and did not increase BDNF whereas BDNF increased after *voluntary* exercise (25,26). Thus, concussed animals, provided they are not forced, appear to appropriately regulate postinjury exercise so that it is beneficial to recovery. Voluntary physical exercise immediately after or within days of traumatic brain injury (TBI) has been shown in other animal studies to promote neuroplasticity (27), increase proliferation of neuronal stem cells (28), reduce neuronal degeneration and apoptotic cell death around the damaged area (29), increase Purkinje neurons and suppress formation of reactive astrocytes (30), and improve cognitive performance in association with reduced DNA fragmentation in the hippocampus (31). In a recent study, Mychasiuk et al. (32) examined the influence of very early exercise (within 1–3 d) on behavior and gene expression (including BDNF) in prefrontal cortex and hippocampus after simulated concussion. Rodents deprived of social interaction and exercise, a combination that recalls recommendations for complete rest until symptom resolution (*i.e.*, "cocooning"), did not recover whereas early voluntary exercise significantly improved motor function and cognition, with the greatest improvement in rats given access to running wheels before injury, that is, the pretrained "rat athletes." This study provides preclinical support for the observation that trained human athletes with PPCS recovered much faster than nonathletes after subthreshold aerobic exercise treatment (33). Thus, animal and human data show that the physically trained brain is different than the sedentary brain and responds positively to controlled exercise after concussion.

### Systematic Evaluation of Exercise Tolerance After Concussion

The Buffalo Concussion Treadmill Test (BCTT) (34) is a systematic and reliable method to determine the symptom-exacerbation exercise threshold in concussed patients (35). The data have been used in adult PPCS patients to prescribe an individualized, progressive subsymptom threshold aerobic exercise program that safely improved symptoms and helped restore function (*i.e.*, return to sport and work) (33,36). Before the test begins the patient is asked to rate his/her symptoms on a Visual Analogue Scale (VAS, 0–10). Then, she or he walks on a level treadmill at 3.2 or 3.6 mph (depending on patient height) at 0-degree incline. The incline is increased by 1 degree after each minute for the first 15 min, and then, the speed is increased 0.4 mph each minute thereafter. Each minute the HR, symptom severity (VAS), and Borg RPE (37) are recorded until symptom exacerbation or voluntary exhaustion, followed by a cool-down period. Voluntary exhaustion is defined as 17 or more (maximum of 20) on the RPE scale. Symptom exacerbation is defined as an increase of 3 points or more from the preexercise VAS value (a point or more for an increase in symptoms and a point for appearance of a new symptom). Patients typically experience increased headache or head pressure, dizziness, or visual symptoms. They are instructed to report and not to push through symptoms and the examiner should observe for visible signs of distress, which may prompt test cessation.

Because the treadmill is not appropriate for all patients, we developed the Buffalo Concussion Bike Test (BCBT). We calculated the oxygen consumption ($\dot{V}O_2$) during treadmill walking using the American College of Sports Medicine (ACSM) Metabolic Equations (38). We then determined, based on the individual's weight (kg), the bike resistance required to achieve an equivalent $\dot{V}O_2$ for each treadmill stage. The patient begins on a stationary bike (that has adjustable resistance) at 60 revolutions per minute. The initial resistance is set and increased every 2 min (according to weight in kg), during which the HR, symptom severity, and RPE are recorded until voluntary exhaustion or symptom exacerbation as with the BCTT. Figure 1 shows that in acutely concussed adolescents ($n$ = 10, 15.7 ± 1.06 years, <7 d from injury) the BCBT was equivalent to the BCTT with respect to rate of HR rise during each stage and, importantly, the HR at symptom exacerbation threshold was equivalent to the BCTT (132 ± 25 vs 137 ± 32 bpm; lower equivalence limit $P$ value = 0.029; upper equivalence limit $P$ value = 0.004). Figure 2 shows the severity of symptoms (VAS) at each stage of the BCTT and BCBT, with 95% CI.

### Exercise Prescription

After establishing the submaximal symptom exacerbation threshold, the patient is prescribed exercise (on a stationary cycle for the first week and then a treadmill) for a minimum of 20 $min \cdot d^{-1}$ at an intensity or "dose" of 80% to 90% of the threshold HR achieved on the exercise test, once per day for 6 to 7 $d \cdot wk^{-1}$ using an HR monitor. The patient is instructed to warm up to the target HR and should have someone present for safety monitoring. The intensity of exercise chosen was based on the principles of safety and the amount of weekly exercise needed to achieve a cardiovascular



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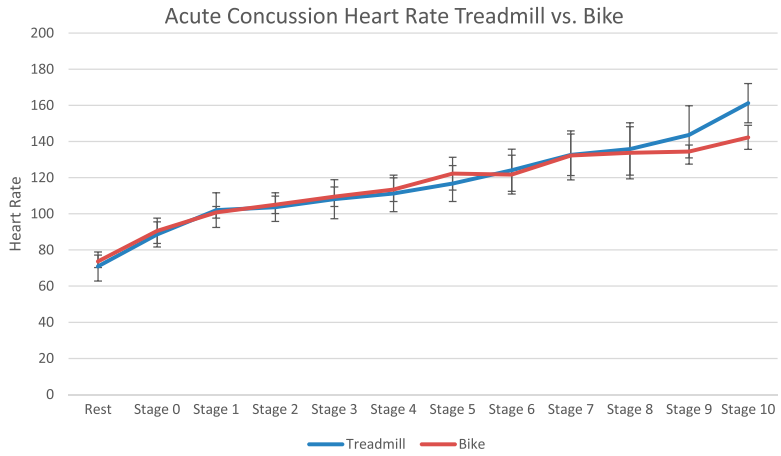


**Figure 1:** HR during stages of the BCTT compared with the BCBT, with 95% confidence interval (CI).

training effect and to modify cardiac autonomic function (39). Exercise is stopped at the first sign of symptom exacerbation, which is defined as a 2 point or more increase from the preexercise baseline symptom level. The BCTT/BCBT can be repeated every 2 to 3 wk to establish a new symptom-limited threshold HR. A more reasonable and cost-effective approach is to increase the HR target by 5 to 10 bpm every 1 to 2 wk, provided the patient is responding favorably (36). Athletes generally respond faster (33) and can increase by 10 bpm every 1 to 2 wk, whereas nonathletes typically respond better to 5 bpm increments every 2 wk. Rate of exercise intensity progression varies and some patients may have to stay at a steady HR for more than 2 wk. The purpose is to give the patient specific goals to achieve without focusing on speed to recovery. Cardiovascular and cerebrovascular physiological recovery is defined as the ability to exercise to voluntary exhaustion at ≥80% of age predicted maximum HR for 20 min several days in a row without symptom exacerbation (33). Patients can then safely begin the Berlin graduated return-to-play (RTP) strategy (1,40). Exercise testing should be considered only for patients without orthopedic or vestibular problems that increase the risk of falling off the treadmill and only in those patients who are at low risk for cardiac disease (33). Recently, we have shown that the BCTT does not increase symptoms the day after testing or delay recovery in adolescents when performed within a week of SRC, provided stopping criteria are followed (4).

### The Physiology of Concussion and Effect of Exercise

A recent systematic review as part of the 2017 CISG consensus meeting summarized the physiological disturbances of SRC (41). The metabolic and physiological changes of concussion result, among other things, in altered function of the autonomic nervous system (ANS) and control of cerebral blood flow (CBF) (42). The primary ANS control center in the brainstem may be damaged in concussion, particularly if there is a rotational force applied to the upper cervical spine (43). Consistent with this, brainstem DTI changes have been reported in patients with PPCS (44). Altered autonomic regulation after TBI is believed to be due to changes in the autonomic centers in the brain and/or an uncoupling of the connections between the central ANS, the arterial baroreceptors, and the heart (45), and studies have shown abnormal ANS function when moving from rest to a state of increased metabolic demand acutely after concussion (46) and in those with PPCS (47).

Of relevance to concussion, the ANS controls the CBF response to exercise (48,49). In a recent controlled study (3), we showed that female college athletes with PPCS had abnormally low sensitivity to the arterial $CO_2$ tension

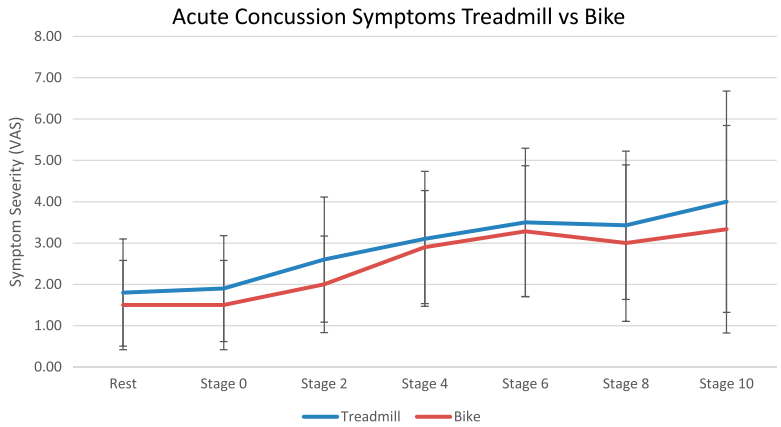


**Figure 2:** Symptom severity at each stage of the BCTT compared with the BCBT, with 95% CI.



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($PaCO_2$) that caused a relative hypoventilation during exercise that raised their $PaCO_2$ levels out of proportion to exercise intensity. Cerebral blood flow is directly proportional to $PaCO_2$; hence, elevated $PaCO_2$ raised exercise CBF disproportionately to exercise intensity and was associated with symptoms of headache and dizziness that significantly reduced exercise tolerance. Subthreshold aerobic exercise treatment increased $CO_2$ sensitivity to normal, which normalized $PaCO_2$, exercise ventilation, CBF, and exercise tolerance, and resolved symptoms. The $CO_2$ sensors are located in the brainstem near the autonomic control centers for cardiopulmonary function. Thus, subthreshold aerobic exercise improved the central physiology of the concussed brain and reduced symptoms in athletes with PPCS. In a different study, we showed that aerobic exercise treatment restored abnormal local CBF regulation to normal and resolved clinical symptoms in PPCS patients, whereas a placebo stretching program did not (50). A recent study using model-based prospective end-tidal $CO_2$ targeting and functional magnetic resonance imaging (fMRI) showed patient-specific alterations in cerebrovascular responsiveness (CVR, the change in CBF in response to a vasodilatory or vasoconstrictive stimulus such as $CO_2$) in the acute and subacute phases of recovery from SRC. A predominant pattern of increased $CO_2$ CVR was seen, including those with exercise intolerance on the BCTT (Mutch et al., 2018 *in press*). These studies suggest that some concussion symptoms and exercise intolerance are related to abnormal ANS regulation of CBF and that individualized, controlled aerobic exercise can restore CBF regulation and exercise tolerance to normal.

Until recently, the traditional therapy for concussion and for PPCS has been rest and avoidance of activity (12). Prolonged rest and social isolation, however, exacerbate symptoms and delay recovery in adolescents (16), results that are similar to preclinical animal models of simulated concussion in rodents (32). Physical deconditioning from prolonged rest can impair autonomic control of CBF (51), whereas exercise training improves CBF control (52) and ANS balance (53). Individualized subthreshold exercise treatment for PPCS patients is safe, nonpharmacological, and well accepted as < 10% of subjects refused exercise treatment (36). An important translational aspect is that the BCTT/BCBT represents a clinical proxy of concussion physiology because exercise intolerance after concussion is associated with abnormal autonomic cardiopulmonary control, whereas restoration of exercise tolerance signals normalization of these fundamental physiological mechanisms. The ability to exercise to exhaustion on a treadmill test without symptom exacerbation is one definition of physiological recovery from concussion (50), which conforms to expert consensus opinion about recovery and readiness to return to activity (54). Normalization of aerobic exercise tolerance may not, however, coincide with recovery of full neurological function after SRC. Recovery of optimal perception-action neurological processing appears to be an equally important criterion for establishing readiness to return to sport because altered gait balance control (55) and an increased risk of musculoskeletal injury have been reported after SRC (56).

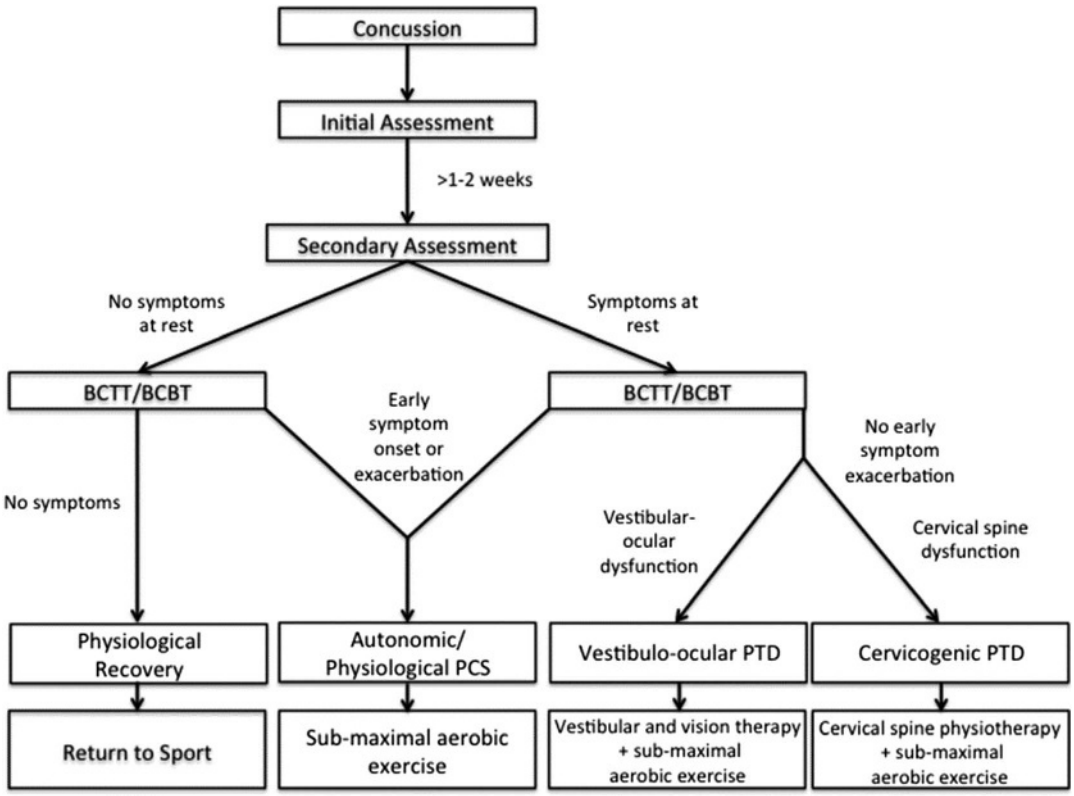


**Figure 3:** Algorithm for diagnosis of physiological postconcussion syndrome (PCS) and different post-traumatic disorders (PTDs) (image adapted from Ellis MJ, Leddy JJ, Willer BS. Physiological, vestibulo-ocular and cervicogenic postconcussion disorders: an evidence-based classification system with directions for treatment. *Brain Injury*. 2015; 29(2):238–48).



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Patient BCTT data can help make the difficult return-to-activity decision for clinicians more objective and physiologically based, as the return of exercise tolerance is a primary determinant of the ability of adolescents to safely return to sport after SRC (40).

### The Differential Diagnosis of PPCS Using Exercise Testing and a Physical Examination

It is clinically useful to classify PPCS patients as having true autonomic/physiological postconcussion syndrome (PCS, defined by exercise intolerance and responsiveness to aerobic exercise treatment) or one of several "posttraumatic disorders" (PTDs, symptoms in the setting of normal exercise tolerance that are not from the metabolic disturbance of concussion (57)) rather than "postconcussion syndrome" because there is more than one cause of PPCS (36,58,59). The patient's performance and symptom pattern during the BCTT/BCBT combined with a pertinent pretest physical examination (60) can help with the differential diagnosis of PPCS (Fig. 3). The PPCS patients who exercise to exhaustion without exacerbation of headache or other symptoms no longer have physiological concussion; rather, they have different symptom-generator(s), most commonly a cervical injury, vestibular/ocular dysfunction, posttraumatic headache syndrome, or a combination (36,61,62). A careful physical examination of the cervical spine and a neurologic examination focusing on the vestibular system and oculomotor responses can help identify sources of symptoms, such as dizziness, headache, trouble concentrating, and blurred vision (58). Exercise itself can induce symptoms of fatigue, headache, and dizziness near voluntary exhaustion, whereas concussed patients are limited by symptoms early (typically at 50% to 70% of age-predicted HR maximum) (4,33). The key differentiating point is that those with cervicogenic headache or cervicogenic dizziness are able to exercise near to exhaustion, despite symptoms, whereas concussed patients stop early at a submaximal level because of significant symptom exacerbation. Some patients with severe (usually peripheral) vestibular dysfunction also will stop exercise tests early, where it is very clear that the vestibular dysfunction is the cause. Many patients with PPCS therefore no longer have concussion as the source of their symptoms (62). They may benefit from aerobic therapy combined with additional targeted interventions tailored to specific posttraumatic disorders or etiologies (63). Thus, the BCTT/BCBT combined with a pertinent physical examination can help the practitioner narrow the differential diagnosis of PPCS and direct therapy to the specific cause, enhancing the RTP process.

### Rest, Moderate Physical Activity, and Prescribed Aerobic Exercise in Concussion Management

The Table shows studies that have evaluated rest and either moderate levels of physical activity or prescribed aerobic exercise in concussion and PPCS. One prospective observational study from a pediatric office showed that, in patients who recovered within 30 d, those prescribed immediate cognitive and physical rest recovered 4.6 d sooner than those with delayed cognitive and physical rest (64), which corresponds with an earlier prospective cohort study that showed increased cognitive activity in the first weeks after concussion was associated with longer recovery (65). Recent prospective studies, including one RCT, however, demonstrated that prescribed strict rest in the acute recovery phase was not as efficacious for symptomatic recovery as unregulated light physical and cognitive activity (16,66). A retrospective study found that adolescent athletes who reported high levels of activity during the subacute phase after SRC (*e.g.,* participation in a sports game) had more symptoms and worse neurocognitive performance than those reporting moderate levels of activity (*e.g.,* slow jogging) (15). In a prospective cohort study, higher

**Table.**
**Studies (and design) that evaluated rest, moderate physical activity, or exercise testing and/or prescribed aerobic exercise in patients shortly after concussion or in those with PPCS.**

| Rest and Moderate Physical Activity | Exercise Testing and/or Prescribed Aerobic Exercise |
|---|---|
| •Majerske et al. (2008) — retrospective cohort | •Gagnon et al. (2009) — Prospective Case Series |
| •Brown et al. (2014) — retrospective cohort | •Leddy et al. (2010) — Prospective Case Series |
| •Thomas et al. (2015) — RCT | •Baker et al. (2012) — Retrospective Cohort |
| •Buckley et al. (2015) — prospective cohort | •Leddy et al. (2013) — Quasi Experimental |
| •Silverberg et al. (2016) — secondary analysis of RCT | •Clausen et al. (2015) — Prospective Cohort |
| | •Maerlender et al. (2015) — RCT |
| | •Dematteo et al. (2015) — Prospective Cohort |
| •Grool et al. (2016) — prospective multicenter cohort | •Cordingley et al. (2016) — Retrospective Cohort |
| •Howell et al. (2016) — prospective cohort | •Gagnon et al. (2016) — Prospective case series |
| | •Kurowski et al. (2017) — RCT |
| •Sufrinko et al. (2017) — secondary analysis of RCT | •Chrisman et al. (2017) — Retrospective Cohort |
| •Taubman et al. (2016) — prospective cohort | •Leddy et al. (2017) — RCT |
| | •Chan et al. (2018) — RCT |

RCT, randomized controlled trial.



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levels of physical activity after injury in those ages 13 to 18 years were associated with shorter symptom duration (67). In a secondary analysis of clinical trial data, an abrupt increase in mental activity (*i.e.*, returning to school and extracurricular activities) increased the risk of a symptom spike, most of which were of short duration (<24 h) (68). In another secondary analysis of the same clinical trial data, patients evaluated in the emergency room who demonstrated certain *signs* of concussion (*e.g.*, confusion or posttraumatic amnesia) benefited from prescribed strict rest, whereas patients with mainly symptoms were more likely to remain symptomatic if prescribed rest (69). In a large prospective, multicenter emergency department study of acute concussion ($n$ = 3063, 5 to 18 years), physical activity reported within 7 d of injury compared with no physical activity was associated with a significantly reduced risk of PPCS at 28 d (24.6% vs 43.5%). Physical activity included light aerobic exercise (32.9%), sport-specific exercise (8.9%), noncontact drills (5.9%), full-contact practice (4.4%), and full competition (17.4%).

The early studies of aerobic exercise treatment were nonrandomized, uncontrolled prospective case series in patients with PPCS (33,70) that showed controlled exercise was safe and effective, whether based upon individual treadmill test performance (33) or a generic submaximal prescription (50% to 60% of estimated maximal capacity) combined with coordination and visualization exercises (70). Leddy et al. (33) showed that the rate of PPCS symptom improvement was related to peak exercise HR, suggesting a physiological effect of aerobic exercise, and that athletes recovered significantly faster than nonathletes (25 ± 8.7 vs 74.8 ± 27.2 d, $P$ = 0.01). In a retrospective review of patients treated with individualized exercise therapy, 72% who had participated in the exercise rehabilitation program returned to full daily functioning at 1 year, including 77% of those who demonstrated exercise intolerance and were considered to have true physiological PCS (36). In a small placebo-controlled quasi-experimental trial, only aerobic exercise-treated PPCS subjects (not placebo stretching subjects) increased exercise tolerance and reduced symptoms in association with normalization of fMRI activation patterns to healthy control levels (50). This small but informative study provided preliminary evidence that some symptoms in PPCS patients may be related to abnormal CBF regulation and that exercise rehabilitation restored normal CBF regulation in association with clinical recovery. In a follow-up prospective controlled experimental study (discussed above), six collegiate female athletes with PPCS had low $CO_2$ sensitivity (ventilatory response to increasing $CO_2$ fraction in the inspired gas, a measure of brainstem physiology) that blunted their exercise ventilation and raised CBF velocity (measured on transcranial Doppler) during treadmill exercise in association with symptom exacerbation and premature exercise cessation (3). Subthreshold exercise treatment over 12 wk normalized their $CO_2$ sensitivity, ventilation, CBF velocity, and exercise tolerance. The data indicate that some athletes with PPCS have exercise intolerance due to abnormal CBF regulation that may be the result of concussion-induced altered sensitivity to $CO_2$ in the brainstem. Return of normal CBF control and of exercise tolerance may therefore be physiological markers of cerebrovascular recovery from concussion.

Recent trials have investigated exercise as both an evaluative tool and as a treatment for concussion and PPCS. Maerlender et al. (71) randomly assigned acutely concussed collegiate athletes to either no structured exercise or to exercise on a stationary cycle at a perceived exertion level of "mild" to "moderate" for 20 $min \cdot d^{-1}$, beginning the day of concussion diagnosis. They found no effect of immediate exercise on time to recovery (exertion group 15 d vs 13 d in controls) and athletes who reported more vigorous exertion tended to take longer to recover. Nevertheless, they concluded that starting mild to moderate exercise very early after injury was safe and that mild symptom increases should not interfere with recovery. In a prospective cross-sectional study, Dematteo et al. (72) evaluated the response of youth with PPCS to the McMaster All-Out Progressive Continuous Cycling Test. The number and severity of symptoms improved significantly in the majority of subjects in the 24 h after the exercise test. They concluded that standardized exertion testing is safe and is important for the evaluation of symptoms and readiness to return to activity, particularly in youth slow to recover. In a retrospective study, Cordingley et al. (73) evaluated the BCTT in pediatric patients with SRC and submaximal aerobic exercise treatment in those with physiological postconcussion disorder (PCD, diagnosed by persistent exercise intolerance). A total of 106 SRC patients (mean age, 15.1 years; range, 11 to 19 years) performed 141 treadmill tests with no serious complications. The BCTT confirmed physiological recovery in 97%, allowing successful return to play in 94%, and helped to diagnose physiological PCD in 58 patients and cervicogenic PCD in 1 patient. Of the 41 patients with physiological PCD who had completed follow-up and were treated with submaximal exercise (and concurrent targeted treatment of vestibulo-ocular and cervical spine dysfunction as indicated), 90% were clinically improved and 81% successfully returned to sport. Patients who did not respond or experienced an incomplete response to exercise included seven with migraines and one with a postinjury psychiatric disorder. The authors concluded that the BCTT was a safe, tolerable, and clinically valuable tool for the evaluation and management of pediatric SRC.

Recent studies demonstrate the safety and efficacy of prescribed aerobic exercise treatment for youth with PPCS. Kurowski et al. (74) randomly assigned 30 adolescents (age, 12 to 17 years) with persistent symptoms for 4 to 16 wk after mild TBI to either subsymptom exacerbation aerobic training or to a full-body stretching (placebo) program. Importantly, they included only subjects with demonstrated exercise intolerance and excluded those with a cervical injury. Despite lower adherence to the home exercise program, there was a greater rate of symptomatic improvement over 6 wk in the aerobic training group versus stretching\placebo, suggesting a physiological effect of aerobic exercise on recovery from PPCS. In a retrospective cohort study of 83 PPCS youth (aged 15 years, 54% female, 76% SRC, symptoms >1 month), Chrisman et al. (75) reported that symptoms decreased exponentially following initiation of prescribed subthreshold exercise and that recovery trajectory did not differ by duration of symptoms at presentation (<6 wk, 6–12, or >12 wk).

## Summary and Conclusions

This article presents emerging evidence for subthreshold aerobic exercise as a non-pharmacological "medicine" to safely evaluate concussion and speed recovery for patients



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with PPCS. The symptom-exacerbation threshold HR can be used to prescribe an individualized subthreshold "dose" of aerobic exercise for a progressive training program that can safely improve symptoms, speed return to activity, and restore function in many patients with PPCS. Systematic evaluation of exercise tolerance combined with a pertinent physical examination is useful for the differential diagnosis of PPCS, which is essential to prescribing aerobic exercise as well as other forms of exercise as medicine to treat PPCS (*e.g.*, cervical, vestibular, and vision therapies). The latest CISG consensus guidelines recommend a more active approach to SRC treatment and there is emerging evidence for the potential effectiveness of controlled aerobic exercise in the *acute phase* after adolescent SRC.

Experimental human studies show that concussion affects ANS control of ventilation, $PaCO_2$ levels, and CBF. Elevated $PaCO_2$ levels and CBF during exercise are associated with symptoms that reduce exercise tolerance in concussed patients. Some experimental and clinical treatment studies suggest a beneficial effect of controlled exercise on this pathophysiology of concussion. Exercise intolerance may therefore prove to be one "physiological biomarker" of concussion whereas normalization of exercise tolerance, combined with recovery of optimal perception–action neurological processing, may prove to be clinically useful biomarkers of recovery and for determining physiological readiness to return to sport after SRC.

Individualized aerobic exercise is a nonpharmaceutical intervention that challenges the old paradigm of prolonged rest, has minimal adverse effects, can be implemented with standard equipment, and could be used at many physician offices and health facilities, including military facilities and in the field, with relative ease. Further research should examine the optimal timing and dose of guided aerobic exercise for the active treatment of concussion, its potential to prevent PPCS, and more thoroughly investigate the physiological and neurophysiological mechanisms for its effect.

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